HERMITIAN–YANG–MILLS FLOW ON DEGENERATING CALABI–YAU THREEFOLDS OF TYPE K

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Abstract. We construct a rank-four holomorphic bundle family on a compact degeneration, with smoothing parameter t, of Enriques Calabi–Yau threefolds. The threefolds are free quotients of products of a K3 surface and an elliptic curve degenerating to a cycle of four rational curves. On the unitary gluing locus, the bundles have nonflat Hermitian–Einstein connections for prescribed product-induced Kähler metrics, which need not be Ricci-flat. We identify all five small eigenvalues of the trace-free endomorphism Laplacian, prove relative asymptotics uniform up to strict polystability, and establish a uniform complementary gap. The effective quadratic energy lies on an interval with one-dimensional endpoint constraints whose relative angle records holonomy around the elliptic central-fibre cycle. We derive a finite-dimensional gradient flow of noncommuting positive Hermitian matrices and reconstruct comparison metrics approximating actual Hermitian–Yang–Mills trajectories in global L^2 affine distance. With determinant fixed to that of the Hermitian–Einstein reference metric, an open class of initial metrics whose relative logarithms are globally bounded and whose component-exterior restrictions admit uniformly bounded $C^{6,\beta}$ extensions to the normalization components, $0 < \beta < 1$, enters this description after physical time $s = o(L)$, where $L = \log(1/|t|)$. Tracking is uniform on bounded s/L intervals and, where the least effective eigenvalue is bounded below by a positive constant, for all subsequent times. Within the specified family, the five eigenvalues, elliptic component areas and known normalization L asymptotically recover the projective conjugacy class of the cycle transport. Infinite physical-time limits need not commute with degeneration. The rescaled limits commute under this gap bound when the selected parallel limits of the component flows and the gluing parameters converge, but can fail near reducibility.

1. Introduction

A degeneration of the base can create slowly relaxing endomorphisms of a Hermitian–Einstein bundle even when its connection has nonzero curvature. The local parallel algebras, their identifications across the degenerating regions, and the symmetries of the compact family determine which of these endomorphisms survive globally. We study this mechanism on a compact Calabi–Yau threefold with a nontrivial rank-four bundle. Its effective metric variables form a five-dimensional constrained space, and the constraints retain gluing information that the scalar dual graph of the central fibre does not record.

Our threefolds are free quotients of a product of a fixed K3 surface and an elliptic curve. Such Calabi–Yau threefolds are called of *type K*. Here a Calabi–Yau threefold means a smooth projective threefold with trivial canonical bundle and $h^{1,0} = h^{2,0} = 0$. The examples have infinite fundamental group and an index-two product cover. The Kähler metrics used in the analysis descend from a fixed Ricci-flat K3 metric and a smooth ambient metric on the

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degenerating elliptic family. They need not be Ricci-flat. The geometric construction, its actual bundle connection, and these metric choices are specified in Section 2.

1.1. **The family and the main result.** Let S be a projective K3 surface with a free Enriques involution ι. Resolve the two successive singular points of

$$y^2z = x(x-z)(x-t^2z)$$

over $t = 0$. This gives a projective elliptic family $\mathcal{E} \to \Delta$ with central fibre a cycle of four rational curves D_j, indexed modulo four. Elliptic inversion σ acts by $j \mapsto -j$. Set $X_t = (S \times E_t)/\langle(\iota, \sigma)\rangle$. The central threefold has three components, with interval dual graph. Choose the K3 metric to have volume one. The actual elliptic component areas are $A_0, A_1, A_2, A_3 = A_1 > 0$.

For $\theta \in \mathbb{R}$ let

$$R = e^{i\theta\sigma_z/4}, \qquad J = i\sigma_x, \qquad U_\theta = R^4, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The rank-two flat bundle V_θ on the elliptic cover has transport R at each oriented node incidence, cycle holonomy U_θ, and trivial vanishing-cycle holonomy. Its lift of σ has square $-I$. Tensoring this lift with $i\,d\iota$ gives a genuine order-two action on $T^{1,0}S \otimes V_\theta$, which descends to a rank-four bundle $\mathcal{B}_{t,\theta}$. Section 2 constructs these transitions on an actual cover of the compact total space and verifies the descent. In particular, the family is not specified merely by a proposed local gluing diagram.

Let $L = \log(1/|t|)$. On $\mathbb{R} \oplus \mathbb{R}^3 \oplus \mathbb{R}$ define the energy and mass

$$\begin{aligned} q_\theta(a, X, b) &= 2\big(\|au - X\|^2 + \|X - bv_\theta\|^2\big), \\ m_A(a, X, b) &= A_0a^2 + 2A_1\|X\|^2 + A_2b^2, \\ u &= (1, 0, 0), \qquad v_\theta = (\cos\theta, \sin\theta, 0). \end{aligned} \tag{1}$$

Write $\mu_0 \le \cdots \le \mu_4$ for their generalized eigenvalues, including zero modes. Let $\mathcal{P}_2$ be the space of positive Hermitian two-by-two matrices of determinant one, with affine invariant metric (26), and let d_2 be its geodesic distance. The finite-dimensional configuration space is

$$\mathcal{F} = \{(H_j) \in \mathcal{P}_2^4 : H_{-j} = JH_jJ^*\}, \qquad g_M = \sum_j A_j g_{H_j}.$$

It is a five-dimensional geodesically convex submanifold. The two endpoint factors are one-dimensional, whereas the middle factor has dimension three and weight $2A_1$.

The initial tuple for the effective evolution is selected from the spatial initial metric. On each fixed normalization component, run the HYM metric flow and take its parallel limiting metric. We call this limit the *component phase*. It is a positive metric in the component's parallel endomorphism algebra, rather than an angular parameter. Theorem 6.1 proves its existence and quantitative dependence on the initial metric. In the degeneration, the component restrictions admit smooth extensions across shrinking node discs; the allowed extensions and their asymptotically vanishing phase ambiguity are specified in Section 7. The norm used there controls the logarithm of the relative metric globally in C^0 and controls extensions of its component-exterior restrictions to the fixed spaces $S \times D_j$ in $C^{6,\beta}$. It permits spatial variation in both factors and arbitrary smooth changes of bounded amplitude inside the middle neck regions.

Theorem 1.1. *For the compact family, bundle and prescribed Kähler metrics just described, the following statements hold on the same smooth fibres.*

(1) *The bundle has a holomorphically trivial determinant and an explicit determinant-normalized Hermitian–Einstein metric h_0. On an embedded K3 slice its second Chern number is 48. It is stable for $\theta \notin \pi\mathbb{Z}$ and strictly polystable for $\theta \in \pi\mathbb{Z}$.*

(2) *Let $\lambda_k(t,\theta)$ be the eigenvalues, indexed from zero, of the actual endomorphism connection Laplacian $D_\theta^* D_\theta$ on trace-free h_0-Hermitian endomorphisms. There are $\varepsilon(t) \to 0$ and $\gamma > 0$, independent of θ, such that*

$$(1-\varepsilon(t))\frac{2\pi}{L}\mu_k(\theta) \le \lambda_k(t,\theta) \le (1+\varepsilon(t))\frac{2\pi}{L}\mu_k(\theta) \quad (0 \le k \le 4), \qquad \lambda_5 \ge \gamma. \tag{2}$$

The kernel is one-dimensional precisely at $\theta \in \pi\mathbb{Z}$. The comparison is relative for every positive weak mode, with no restriction on how quickly θ approaches that locus.

(3) *The effective evolution is the negative gradient on $(\mathcal{F}, g_M)$ of*

$$\frac{\pi}{L}\sum_{j \bmod 4} d_2(H_{j+1}, RH_jR^*)^2. \tag{3}$$

For every fixed $R_0 > 0$ and $0 < \beta < 1$, the nonempty open class $\mathcal{B}_{R_0}(t,\theta)$ defined by the component-extension norm (59) has a selected tuple of component phases in $\mathcal{F}$. The class includes non-tensor rank-four metrics. There is a common time $s_*(t) \to \infty$, with $s_*/L \to 0$, after which the actual equation

$$h^{-1}\partial_s h = -2i\Lambda_{\omega_t} F_h$$

is approximated by the reconstructed effective evolution in global squared L^2 affine distance. The error tends to zero uniformly over this class and all θ on each interval $[s_*, s_* + TL]$ with fixed T. If $\mu_0(\theta) \ge \kappa > 0$, the error tends to zero uniformly for every $s \ge s_*$. The reconstruction and error bounds are given in Theorems 5.1, 5.2 and 7.2.

(4) *Suppose the family, L and the three areas are known, but θ and the transition matrices are unknown. From the five actual eigenvalues, the quantity*

$$\frac{A_0A_1^3A_2}{8}\left(\frac{L}{2\pi}\right)^5 \prod_{k=0}^{4}\lambda_k(t,\theta) \tag{4}$$

recovers $\sin^2\theta$ with relative error tending uniformly to zero when it is positive, and equals zero at reducibility. It therefore recovers the conjugacy class of $\mathrm{Ad}(U_\theta)$ in $\mathrm{SO}(3)$ uniformly. The quantitative noise bounds are those of Theorem 8.1.

Both geometric parameters are fixed along every flow trajectory.

Proof. Propositions 2.1–2.5 prove part 1 on the stated family. Theorem 3.1 gives part 2 after the exact K3 product decomposition and the equivariant capacity comparison. Proposition 4.1 derives the field in part 3. Theorems 5.1 and 5.2 prove its reconstruction and comparison with arbitrary matching rank-four initial metrics. The fixed-bundle phase theorem and the preparation estimate of Theorem 7.2 supply such matching from the declared open class. Theorem 8.1 proves part 4. All steps use the same component areas, incidence transport and determinant normalization. The cancellation of the rank-two tangent factor with the degree-two quotient in (21) and (44) ensures that the spectral and nonlinear coefficients agree. □

1.2. **Proof mechanisms and scope.** The exact product connection separates endomorphisms parallel in the K3 direction from a complement with a fixed positive gap. The remaining curve modes are subject to the involution. In the central quotient, this makes the two endpoint parallel spaces lines in a three-dimensional matrix space. Their relative angle records the transport around the cycle upstairs, even though the downstairs dual graph is an interval. Component averaging and neck interpolation respect the involution. The lower spectral bound is controlled by the section's own energy, which preserves relative accuracy when the smallest effective eigenvalue tends arbitrarily rapidly to zero.

For the nonlinear statement, a component Green correction cancels the leading effective velocity. Corrected endpoint geodesics and relative-logarithm cutoffs then give a global Chern residual of order $(1+\log L)^2/L^2$ in integral, uniformly for all physical times. Tensor curvature and descent produce a comparison metric on the actual rank-four bundle. An intrinsic trace inequality compares it to exact solutions with no tensor restriction. To obtain the open class, a separate preparation metric uses spatial component flows. Its node profile has nonzero transverse mean curvature; the proof controls that term using the small node volume. This distinguishes the preparation estimate from a formal extension of the one-dimensional equation.

The all-time assertion requires the positive-rate class in the theorem. The weakest linear time scale near reducibility is instead L/ϑ^2, where ϑ is the distance to $\pi\mathbb{Z}$. Section 8 proves that physical time and degeneration generally do not commute, that rescaled time restores commutation on a positive-rate class, and that approach to reducibility can destroy that commutation again. For spatial initial data, a limit at fixed positive rescaled time also requires convergence of the selected phases. The inverse in the theorem concerns projective bundle transport; it does not recover an integral cohomology ring, a central lift, or a Ricci-flat metric on the threefold.

1.3. **Relation to previous work.** The smooth Enriques Calabi–Yau quotient is classical; see Hashimoto–Kanazawa [HK17] for its place among threefolds of type K. Rank-four bundles on this very quotient also predate the present construction. Thomas [Tho00, Section 2, Theorem 2.4] constructs polystable rank-four bundles with trivial determinant; we use the corrected preprint for the associated statements. Andreas–Curio [AC07, Section 4, Proposition 4.2] give a stability criterion for already non-split extensions on the Enriques elliptic threefold, under their numerical hypotheses. Neither the quotient geometry nor the existence of rank-four bundles on it is the contribution claimed here. Our construction provides a specified nonflat tangent-tensor bundle family for which the degeneration, complete weak spectrum, actual nonlinear evolution and projective inverse can be proved together.

Graph descriptions of small eigenvalues have a substantial history, including Burger's relative spectral comparison for degenerating Riemann surfaces [Bur90]. Yang–Mills flow under degeneration of the surface was studied by Daskalopoulos–Wentworth [DW00]. The ambient-induced metric used here has collapsing neck area; its zero-form mass must be computed for that metric even though curve Dirichlet energy is conformally invariant. Effective HYM dynamics and bundle gluing for this type of curve degeneration are developed in [Wue26a, Wue26b]. We give the required equivariant estimates here and prove their relation to the actual nonflat threefold operator and flow.

The fixed-base existence and order comparison are the classical compact Kähler inputs recalled in [HKKP18, Section 4.1]; the trace comparison is used in the form of [Wan08, Proposition 2.4]. These inputs are valid in arbitrary complex dimension. They do not supply uniform derivative or heat kernel estimates for our degenerating family. The relative weak

spectral comparison, the global residual, the spatial preparation and their parameter/time domains are proved below.

2. The compact family and its rank-four bundle

We first construct the degeneration and its bundle on the entire compact family. The smoothing parameter and the unitary gluing parameter will vary independently. Both are fixed during each metric-flow trajectory.

2.1. **An elliptic family with four central components.** Let Δ be a disc about zero of radius less than one. In $\mathbb{P}^2 \times \Delta$ consider the family

$$\mathcal{W} = \{y^2 z = x(x-z)(x-t^2 z)\}. \tag{5}$$

Its section $[0:1:0]$ supplies the origin on every smooth elliptic fibre. The involution is $\sigma(x,y,z,t) = (x,-y,z,t)$.

Proposition 2.1. *Two successive blowups at singular points over $t=0$ give a proper projective flat family $\mathcal{E} \to \Delta$ with smooth total surface. Its noncentral fibres are smooth elliptic curves. Its central fibre is a cycle $D_0 \cup D_1 \cup D_2 \cup D_3$ of smooth rational curves, with local equation $uv = t$ at each of its four nodes. The involution lifts to $\mathcal{E}$, acts by $D_j \mapsto D_{-j}$, and fixes no node. Indices are taken modulo four.*

Proof. For $t \neq 0$ the roots $0, 1, t^2$ are distinct, so the fibre is a smooth plane cubic. In the chart $y = 1$, the derivative with respect to z of the defining equation is one at $x = z = 0$, proving smoothness at the section. In the chart $z = 1$, write $f = y^2 - x(x-1)(x-t^2)$. The identities $f_y = 2y$ and $f_t = 2tx(x-1)$ show that the only singular point of the total surface over the chosen disc is $(x,y,t) = (0,0,0)$. At $t = 0$ this also follows directly from $f = f_x = f_y = 0$.

Choose the square root of $1 - x$ equal to one at zero and set

$$Y = \frac{y}{\sqrt{1-x}}, \qquad u = 2(x - t^2/2 + iY), \qquad v = 2(x - t^2/2 - iY). \tag{6}$$

The Jacobian in (x,y) at the origin is invertible, and the equation becomes exactly $uv = t^4$. The involution exchanges u and v.

Perform the blowups first on the algebraic surface over $\mathbb{A}^1$ and then restrict to Δ. In the t-chart of the first blowup, $u = tu_1$, $v = tv_1$, and the strict transform is $u_1 v_1 = t^2$. Its only singular point is its origin. In the u-chart, write $v = uv_1$, $t = ut_1$. The equation is $v_1 = u^2 t_1^4$, a smooth chart with coordinates (u, t_1) and base function $t = ut_1$. The v-chart is symmetric. Blow up the remaining origin. Its t-chart has $u_1 = ta$, $v_1 = tb$ and $ab = 1$. Its u_1-chart has $v_1 = u_1 b$, $t = u_1 a$ and $b = a^2$, with base function $t = u_1 a$; the other chart is symmetric. All charts are smooth and the zero divisor of t is reduced with ordinary crossings.

The four crossing charts can be written uniformly as

$$u = x_i^{i+1} y_i^i, \qquad v = x_i^{3-i} y_i^{4-i}, \qquad t = x_i y_i, \qquad 0 \leq i \leq 3, \tag{7}$$

with adjacent transition $x_{i+1} = y_i^{-1}$, $y_{i+1} = x_i y_i^2$ wherever $y_i \neq 0$. They exhibit the chain of three exceptional rational curves. The original central cubic is irreducible and nodal, with rational normalization. Its proper transform joins the two ends of the chain at its two distinct branches. This gives the asserted four-cycle. It is the usual resolution pattern of the A_3 singularity; the charts additionally verify the multiplicities of this particular base function.

Both singular-point centres are invariant under σ, so the universal property of the blowup lifts the involution at each step. In (7) it sends (x_i, y_i) to (y_i, x_i) in chart $3-i$. Label the proper transform by D_0 and the exceptional chain successively by D_1, D_2, D_3. The resulting component action is $j \mapsto -j$. It fixes D_0, D_2 as sets, interchanging their incident nodes, and exchanges D_1, D_3. Thus all four nodes are paired.

Blowups and the original plane-cubic morphism are projective and proper. The resolved total space is smooth, and its nonconstant base coordinate is a non-zero-divisor. It is therefore flat over the nonsingular curve base. The blowups do not change any noncentral fibre. Shrinking the disc if necessary gives the claimed family. □

Choose a smooth Kähler form on $\mathcal{E}$, using a relative projective embedding and a positive base form, and average it under σ. Write the resulting invariant form as $\omega_{\mathcal{E}}$ and put

$$A_j = \int_{D_j} \omega_{\mathcal{E}} > 0, \qquad A_3 = A_1. \tag{8}$$

No equality between A_0, A_1, A_2 is imposed. These actual areas will enter the spectral mass and the finite-dimensional metric evolution.

2.2. **A holomorphic gluing family and its lift.** Choose disjoint node neighbourhoods N_j at $D_j \cap D_{j+1}$ and component neighbourhoods O_j extending slightly into their incidence collars. They can be chosen so that $\sigma(O_j) = O_{-j}$ and $\sigma(N_j) = N_{-j-1}$. Each node set meets its two component sets on separate connected collars, and there are no triple overlaps. For example, choose two radii $0 < a < b$ and restrict to $|t| < a^2$: the two collars of a node are disjoint because $uv = t$. Properness ensures that this cover of a neighbourhood of the compact central fibre contains every fibre over a sufficiently small disc.

For $q \in \mathbb{C}^*$ define

$$R(q) = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_x. \tag{9}$$

Here σ_x is the first Pauli matrix. On $O_j \times \mathbb{C}^*$ and $N_j \times \mathbb{C}^*$ take trivial rank-two bundles, with fibre coordinates w_j and z_j, respectively. Use transitions

$$w_j = z_j, \qquad w_{j+1} = R(q) z_j \tag{10}$$

on the two incidences.

Proposition 2.2. *The transitions (10) define a holomorphic rank-two bundle V with trivialized determinant on $\mathcal{E} \times \mathbb{C}^*$. For fixed q it carries a flat holomorphic connection. On $|q| = 1$ the identity metrics glue to a parallel unitary metric. The maps*

$$w_{-j} = J w_j, \qquad z_{-j-1} = R(q)^{-1} J z_j \tag{11}$$

give a lift of σ whose square is $-I$. They preserve the connection, and on the unit circle preserve its metric.

Proof. The transitions are holomorphic and invertible on the actual overlaps, have determinant one, and have no triple-overlap condition to satisfy. For each fixed q, they are constant in the $\mathcal{E}$ coordinates, so the connections d in the patch frames agree. Flatness in the q direction is not asserted. Unitarity gives the metric statement.

The identities $JR = R^{-1}J$ and $J^2 = -I$ verify the lift. The source left incidence maps to the target right incidence, where $Rz_{-j-1} = Jz_j = Jw_j = w_{-j}$. On the other incidence, $Jw_{j+1} = JRz_j = R^{-1}Jz_j$, as required. Both component and node lifts square to $-I$, since

$(R^{-1}J)^2 = -I$. The compatibility is holomorphic in q and preserves the flat connection for fixed q. □

For the unitary family write

$$q = e^{i\theta/4}, \qquad R = e^{i\theta\sigma_z/4}, \qquad U_\theta = R^4 = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{12}$$

A positively oriented path around the component cycle passes from the j frame to the $j+1$ frame by R at every step. Its actual flat holonomy is therefore U_θ. The vanishing loop has trivial holonomy, because the whole node patch carries the connection d. These loops generate the elliptic fibre's fundamental group. Thus the transition product has been identified with connection holonomy, with an orientation and path specified.

2.3. **The free quotient and its metric.** Fix a projective K3 surface S with a free Enriques involution ι. Such projective examples exist, and the involution acts as -1 on the holomorphic two-form; see [HK17, Example 2.10]. The smooth quotient construction below is the classical Enriques Calabi–Yau example [HK17, Example 3.5]. Choose an invariant Kähler class by averaging an ample class. By [Yau78, Theorem 2], its unique Ricci-flat representative is ι-invariant. Rescale it to obtain a form ω_S with

$$\int_S \frac{\omega_S^2}{2} = 1. \tag{13}$$

The Chern connection on $T^{1,0}S$ has holonomy SU(2) by [Bea85, Propositions 1 and 4]. Its complex parallel endomorphisms are consequently scalar by Schur's lemma. This statement concerns complex-linear tangent endomorphisms, not the larger real tangent commutant. Fix the holomorphic two-form on S to have pointwise norm one for this metric. Such a scaling is possible because its norm is constant for a Ricci-flat Kähler metric. Its dual therefore gives a unit-norm trivialization of $\det T^{1,0}S$.

Proposition 2.3. *The quotient*

$$\mathcal{X} = (S \times \mathcal{E})/\langle\tau\rangle, \qquad \tau = (\iota, \sigma), \tag{14}$$

is a proper projective flat family with smooth total space. Its central fibre is a reduced simple normal-crossing threefold with three smooth components and interval dual graph. For $t \neq 0$, its smooth connected fibre X_t has trivial canonical bundle, $h^{1,0} = h^{2,0} = 0$, and an index-two subgroup $\mathbb{Z}^2$ in its fundamental group. The form $\omega_S + \omega_{\mathcal{E}}|_{E_t}$ descends to a Kähler form ω_t on X_t, where E_t is the elliptic fibre.

Proof. The involution is free on the entire total space because ι is free. The quotient map is finite *étale*; hence the quotient is smooth and the local crossing equation is unchanged. The fixed component sets give $(S \times D_0)/\langle\tau\rangle$ and $(S \times D_2)/\langle\tau\rangle$. The exchanged pair $S \times D_1, S \times D_3$ has image isomorphic to $S \times \mathbb{P}^1$. The two pairs of nodes give two smooth double loci isomorphic to S. No node is fixed and no branch of a node is identified with the other branch. These observations prove the asserted simple normal crossings and interval incidence.

For projectivity choose a relatively ample line bundle L on $S \times \mathcal{E}$, using an ample line on S and a relative embedding of $\mathcal{E}$. The line $L \otimes \tau^* L$ has the order-two linearization which exchanges its factors. It descends under the free action; its descent is relatively ample because ampleness can be tested after finite surjective pullback. Properness follows from the finite quotient of the proper family. Flatness follows from the non-zero-divisor t on the smooth total space.

On each smooth fibre both the K3 two-form and the elliptic one-form change sign under their respective involutions. Their product descends to a nowhere vanishing three-form. Künneth and invariants under the finite *étale* cover give $h^{1,0} = h^{2,0} = 0$. The cover $S \times E_t$ has fundamental group $\mathbb{Z}^2$, which injects with index two into the quotient fundamental group. The product Kähler form is invariant and hence descends. □

The elliptic metric is the restriction of a smooth total-space metric. It is generally not flat, so the descended threefold metric is generally not Ricci-flat. All subsequent analytic statements use this particular metric, including its actual component areas (8).

2.4. **Rank-four descent, determinant and second Chern class.** On $S \times \mathcal{E} \times \mathbb{C}^*$ form

$$\widetilde{\mathcal{B}} = p_S^* T^{1,0} S \otimes p_{\mathcal{E}}^* V. \tag{15}$$

The tangent map $i\,d\iota$ has square $-I$. Tensor it with the lift (11) of σ.

Proposition 2.4. *The tensor lift defines an order-two equivariant structure on $\widetilde{\mathcal{B}}$ and gives a holomorphic rank-four bundle $\mathcal{B}$ on $\mathcal{X} \times \mathbb{C}^*$. Its determinant has an actual holomorphic trivialization. For every $t \neq 0$ and every q, a specified embedded K3 slice $S_e \subset X_t$ satisfies*

$$\int_{S_e} c_2(\mathcal{B}_{t,q}|_{S_e}) = 48. \tag{16}$$

Proof. The two factors of the lift square to $-I$, so their tensor product squares to I. Incidence compatibility follows from Proposition 2.2. Thus the tensor bundle has genuine descent under the free involution, holomorphically in q.

Trivialize $\det T^{1,0}S$ by the dual of its holomorphic two-form. Anti-symplecticity gives $\det(d\iota) = -1$ in these trivializations, and hence $\det(i\,d\iota) = 1$. The matrices J and $R^{-1}J$ have determinant one. Since both tensor factors have rank two,

$$\det(A \otimes B) = (\det A)^2 (\det B)^2 = 1.$$

The determinant trivialization is invariant under the descent datum and therefore descends; this is stronger than a torsion determinant line.

Choose $e \in E_t$ which is not a two-torsion point. The image of $S \times \{e\}$ is an embedded K3 slice: no two of its points are τ-related because $-e \neq e$, and the proper map is locally an embedding under the *étale* quotient. Restriction of the bundle to that slice is $T^{1,0}S \oplus T^{1,0}S$. The identities $c_1(T^{1,0}S) = 0$ and $\int_S c_2(T^{1,0}S) = 24$ give (16). No division of an integral class by the cover degree is used. □

2.5. **The Hermitian–Einstein connection and stability.** Let h_S be the tangent metric induced by ω_S. On the unitary locus equip (15) with $h_S \otimes I$. The lift is unitary, so this metric and its Chern connection descend. Write their descents as h_0 and D_θ.

Proposition 2.5. *For the metric ω_t of Proposition 2.3, the reference metric h_0 is determinant-normalized Hermitian–Einstein with zero Einstein constant. The bundle $\mathcal{B}_{t,\theta}$ is slope stable for $\theta \notin \pi\mathbb{Z}$ and strictly polystable for $\theta \in \pi\mathbb{Z}$. Its complex parallel endomorphism algebra is*

$$\mathrm{End}_{\mathrm{par}}(\mathcal{B}_{t,\theta}) \simeq \mathrm{Comm}(U_\theta) \cap \mathrm{Comm}(J). \tag{17}$$

Proof. The chosen determinant trivialization has unit norm for h_S, and the flat multiplicity determinant has unit norm for I. Hence their tensor determinant, and its invariant descent,

have unit norm for h_0. The product connection has curvature $F_{T^{1,0}S} \otimes I$, since the elliptic factor is flat. Product contraction gives

$$i\Lambda_{\omega_S+\omega_{\mathcal{E}}} F_{\widetilde{\mathcal{B}}} = (i\Lambda_{\omega_S} F_{T^{1,0}S}) \otimes I = 0.$$

This uses Ricci flatness of the K3 metric and imposes no flatness condition on the elliptic metric. It descends to the stated connection on X_t.

The K3 holonomy makes every parallel complex endomorphism on the cover equal to $I_{T^{1,0}S} \otimes M$ for $M \in \mathrm{Comm}(U_\theta)$. Compatibility with the lift at the D_0 component is exactly $MJ = JM$. This proves (17); the other component and node conditions follow by transport and the verified lift.

For $\theta \notin \pi\mathbb{Z}$, U_θ has distinct eigenvalues. Commutation with it makes M diagonal, and commutation with J makes its diagonal entries equal. The parallel algebra is scalar. The Hermitian–Einstein polystable decomposition [Kob87, Theorem 5.8.3] therefore has only one stable summand, because any further summand would give a nonscalar parallel projection. Thus the bundle is stable. At $\theta \in \pi\mathbb{Z}$ the matrix U_θ is central and $\mathrm{Comm}(J) \simeq \mathbb{C} \oplus \mathbb{C}$. Its eigenprojections split the bundle into two rank-two Hermitian–Einstein summands, proving strict polystability. Their slopes are equal and the total slope is zero for the actual ω_t. □

The holomorphic family is defined for $q \in \mathbb{C}^*$, whereas the displayed Hermitian–Einstein connections and stability analysis concern $|q| = 1$. The approach to strict polystability is a genuine change of holomorphic bundle gluing, independent of t. No nonunitary or unstable locus is asserted by this construction.

3. Small eigenvalues of the endomorphism Laplacian

All operators in this section are evaluated at the reference metric h_0. Put $L = \log(1/|t|)$ and let

$$\mathcal{L}_{t,\theta} = D_\theta^* D_\theta \quad \text{on } \mathrm{Herm}_0(\mathcal{B}_{t,\theta}).$$

The subscript zero denotes trace-free, h_0-Hermitian endomorphisms. We use the unnormalized trace norm and the nonnegative connection Laplacian. This operator is the negative infinitesimal generator of determinant-normalized HYM metric flow at h_0. Indeed expansion of the Chern metric equation at $h_0 e^\Psi$ gives $\Psi_s = -D_\theta^* D_\theta \Psi$ to first order; the contracted background curvature commutator vanishes because $i\Lambda F_{D_\theta} = 0$. The operator is not a Yang–Mills Hessian on connection one-forms.

Write $e_x = \sigma_x/\sqrt{2}$, $e_y = \sigma_y/\sqrt{2}$ and $e_z = \sigma_z/\sqrt{2}$, where $\sigma_y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$. These form an orthonormal real basis of $\mathrm{Herm}_0(2)$. Define

$$\begin{aligned} q_\theta(a, X, b) &= 2\big(\|au - X\|^2 + \|X - bv_\theta\|^2\big), \\ m_A(a, X, b) &= A_0 a^2 + 2A_1 \|X\|^2 + A_2 b^2, \\ u = (1,0,0), &\qquad v_\theta = (\cos\theta, \sin\theta, 0), \end{aligned} \tag{18}$$

on $\mathbb{R} \oplus \mathbb{R}^3 \oplus \mathbb{R}$. Denote their generalized eigenvalues by $0 \le \mu_0(\theta) \le \cdots \le \mu_4(\theta)$, including zeros.

Theorem 3.1. *There are $\gamma > 0$ and $\varepsilon(t) \to 0$ as $t \to 0$ such that the eigenvalues of $\mathcal{L}_{t,\theta}$ satisfy*

$$(1 - \varepsilon(t))\frac{2\pi}{L}\mu_k(\theta) \le \lambda_k(t,\theta) \le (1 + \varepsilon(t))\frac{2\pi}{L}\mu_k(\theta) \quad (0 \le k \le 4), \qquad \lambda_5(t,\theta) \ge \gamma. \tag{19}$$

The estimates are uniform in the unitary parameter. For small t there are exactly five eigenvalues below a fixed positive threshold, counting zero modes. The kernel has dimension one when $\theta \in \pi\mathbb{Z}$ and zero otherwise. No relation between t and the distance of θ to $\pi\mathbb{Z}$ is required.

The spectral reduction uses an exact product decomposition followed by an equivariant curve estimate. The first step removes every mode which is nonparallel in the K3 direction. The second step must preserve the fixed-point constraints at the two ends of the quotient interval.

3.1. **The K3 complement and the quotient normalization.** On the cover the bundle connection is $D_S \otimes I + I \otimes D_V$. A product orthonormal frame gives the exact sum decomposition of its endomorphism connection Laplacian. There are no mixed terms in the energy $\int |D\Phi|^2$. Let Π_S be projection onto the parallel tangent endomorphisms in the S variable. By (13) it is

$$\Pi_S \Phi = I_{T^{1,0}S} \otimes \frac{1}{2} \int_S \operatorname{tr}_{T^{1,0}S} \Phi \, dV_S. \tag{20}$$

The resulting multiplicity endomorphism is Hermitian and trace-free when Φ is. The orthogonal complement includes nonconstant scalar functions on S as well as nonscalar tangent endomorphisms. The fixed compact K3 connection Laplacian has a positive gap γ_S on this entire complement. Apply its inequality at every elliptic point and integrate. The elliptic energy is nonnegative, so the product operator is bounded below by γ_S on $\ker \Pi_S$, independently of t and θ.

The deck action preserves the metric and connection and commutes with Π_S. The quotient operator is the restriction to invariant sections; both norm and energy on the cover are divided by two. On the tangent-scalar sector this division cancels the rank-two tangent factor:

$$\|I_{T^{1,0}S} \otimes f\|^2_{L^2(X_t)} = \frac{1}{2} \|I_{T^{1,0}S}\|^2 \operatorname{Vol}(S) \int_{E_t} \|f\|^2 \, dA = \int_{E_t} \|f\|^2 \, dA. \tag{21}$$

The same identity holds for the energy. It remains to determine the curve spectrum in this invariant trace-free multiplicity sector.

3.2. **The five-dimensional invariant space.** In component frames an invariant tuple obeys

$$X_{-j} = \operatorname{Ad}(J) X_j. \tag{22}$$

The fixed subspace of $\operatorname{Ad}(J)$ in $\operatorname{Herm}_0(2)$ is $\mathbb{R} e_x$. Thus $X_0 = a e_x$, $X_2 = b e_x$, and $X_3 = \operatorname{Ad}(J) X_1$. There are exactly five real degrees of freedom, with mass $A_0 a^2 + 2A_1 \|X_1\|^2 + A_2 b^2$.

The transported four-cycle energy is

$$q(X_\bullet) = \sum_{j \bmod 4} \|X_{j+1} - \operatorname{Ad}(R) X_j\|^2.$$

Using $JR = R^{-1}J$ and (22), its opposite terms are equal, giving

$$q(X_\bullet) = 2\big(\|X_1 - \operatorname{Ad}(R) X_0\|^2 + \|X_2 - \operatorname{Ad}(R) X_1\|^2\big).$$

Set $X = \operatorname{Ad}(R^{-1}) X_1$. This orthogonal change of middle coordinates turns the endpoint directions into $u = e_x$ and $v_\theta = \operatorname{Ad}(R^{-2}) e_x = (\cos\theta, \sin\theta, 0)$. It gives exactly (18). In particular, the interval obtained by quotienting the cycle has endpoint *lines* whose relative angle depends on the gluing. Replacing it by a scalar interval Laplacian would discard that information.

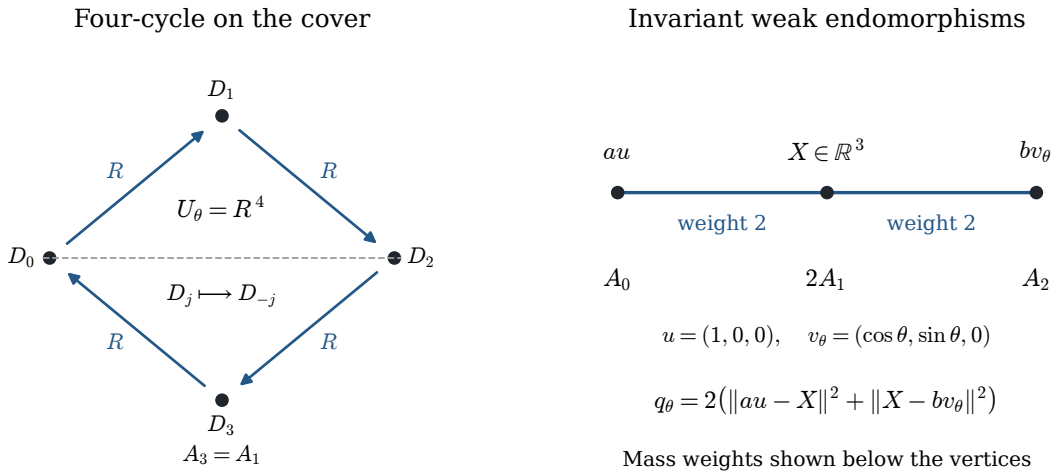


FIGURE 1. The four-cycle on the elliptic cover and its invariant endomorphism problem. After an orthogonal change of middle frame, the endpoint lines are $\mathbb{R}u$ and $\mathbb{R}v_\theta$. The two edge coefficients are 2, and the mass weights are $A_0, 2A_1, A_2$. The diagram represents the exact effective forms (18), not a numerical approximation of the HYM flow.

3.3. **Equivariant capacity and mass estimates.** We record the curve estimates needed for the restriction to invariant sections. Their constants concern the fixed elliptic family and a finite number of component frames, and are uniform in the unitary transitions. For a fixed small cutoff radius a, take paired node discs and invariant component exteriors. A section with vertex values X_j is extended by affine interpolation in $\log|u|$ across a node annulus. Conformal invariance of two-dimensional Dirichlet energy gives

$$E(J_tX) = \frac{2\pi}{L + 2\log a}\, q(X), \qquad \big|\|J_tX\|^2 - m_A(X)\big| \le (Ca^2 + o_a(1))m_A(X). \tag{23}$$

Here $E(f) = \int_{E_t} |D_V f|^2\, dA$, and $o_a(1)$ tends to zero as $t \to 0$ for fixed a. The mass error follows from convergence on the component exteriors and the node-area bound $O(a^2)$.

Both trial interpolation and component averaging commute with the involution. At a reflected node the logarithmic coordinate reverses, and the two endpoints are transported by the verified unitary lift. On a fixed component, averaging commutes with its isometry and the constant frame map J. Consequently (23) restricts to the five-dimensional space (22).

For a section f, let $b_j(f)$ be its mean on a fixed truncated component. Poincaré and trace inequalities on that fixed component bound the difference between $b_j(f)$ and the circular mean at an incident boundary by $C_aE(f)^{1/2}$. Angular averaging and radial Cauchy–Schwarz on the annulus then give

$$q(b(f))^{1/2} \le \left(\sqrt{\frac{L + 2\log a}{2\pi}} + C_a\right) E(f)^{1/2}. \tag{24}$$

This is an estimate by the section's own energy, not an additive error which could dominate a small positive eigenvalue.

For completeness, the mass estimate excluding extra weak modes is

$$\left|m_A(b(f)) - \|f\|^2\right| \le (Ca^2 + C_a\sqrt{\Lambda} + o_a(1))\|f\|^2 \tag{25}$$

on a span of at most six eigensections with eigenvalues at most $\Lambda \le 1$. On fixed component collars, Poincaré and interior elliptic estimates bound the boundary oscillation around the component mean by $C_a\sqrt{\Lambda}$ for an orthonormal eigensection. Their constants are uniform because the local flat equation and metric have uniform fixed-chart bounds. The corresponding statement for a combination of at most six eigensections follows by Cauchy–Schwarz.

On each annulus let v be the harmonic section with the same boundary values as f. The maximum principle controls v by those boundary values, and its mass is $O(a^2)$ times their squared bound. The difference $f - v$ has zero boundary values and satisfies

$$\int_{\text{annulus}} |f - v|^2 \, dA \le Ca^2 \int_{\text{annulus}} |D_V(f - v)|^2 \, dA \le Ca^2 E(f).$$

To obtain the first inequality uniformly in the annulus modulus, use $F = |u|^2 + |v|^2$ on the Euclidean curve $uv = t$. Its divergence Laplacian is four and $|\nabla F| \le Ca$. Integration by parts against the squared norm of a zero-boundary section and Cauchy–Schwarz prove the inequality. The actual induced metric is uniformly comparable to this metric, and Dirichlet energy is conformally invariant. The second inequality follows from the Dirichlet orthogonality of v and $f - v$. Summing these bounds with the component mass estimate proves (25). Invariant sections have invariant means, so this estimate also restricts to the required sector.

Proof of Theorem 3.1. The invariant trial map and min–max give five eigenvalues of order $O(L^{-1})$ and their relative upper bounds. If a sixth invariant curve eigenvalue tended to zero, choose a small and then t small. Equation (25) would make the averaging map injective from a six-dimensional spectral space into the five-dimensional invariant tuple space, a contradiction. The invariant curve complement therefore has a uniform positive gap γ_E.

Apply (24) and (25) on each initial low spectral subspace and use finite-dimensional min–max. Along with (23), they give the relative lower and upper bounds. First fix a for a prescribed error and then take t small. This produces one function $\varepsilon(t) \to 0$ uniform in θ; no small effective eigenvalue is divided out in any estimate.

The product decomposition and (21) transfer these five eigenvalues to the threefold with the same coefficient. Every other mode has lower bound $\gamma = \min\{\gamma_S, \gamma_E\}$ after reducing it if necessary. Finally zero energy means a compatible parallel tuple. In (18) this is the equation $X = au = bv_\theta$, which has one-dimensional solution space when $\sin\theta = 0$ and only the zero solution otherwise. It agrees with (17), proving the exact kernel assertion. □

Corollary 3.2. *Let $\Pi_{t,\theta}$ be the orthogonal projection onto the five weak eigenspaces, counting their exact kernel. On its orthogonal complement,*

$$\|\mathcal{L}_{t,\theta}^{-1}(I - \Pi_{t,\theta})\|_{L^2 \to L^2} \le \gamma^{-1}, \qquad \|e^{-s\mathcal{L}_{t,\theta}}(I - \Pi_{t,\theta})\|_{L^2 \to L^2} \le e^{-\gamma s}.$$

For $z \notin [\gamma, \infty)$, the complementary resolvent has norm at most $\operatorname{dist}(z, [\gamma, \infty))^{-1}$.

Proof. Apply the spectral theorem to the nonnegative self-adjoint operator on the complementary subspace. □

These are L^2 estimates in the actual bundle metric. They do not assert uniform global Schauder bounds across the degenerating necks.

4. The constrained finite-dimensional metric evolution

Let $\mathcal{P}_r = \{H = H^* > 0 : \det H = 1\}$. We use its affine invariant metric and distance

$$g_H(V,W) = \operatorname{tr}(H^{-1}VH^{-1}W), \qquad d_r(H,K)^2 = \operatorname{tr}\Big(\log(H^{-1/2}KH^{-1/2})^2\Big). \tag{26}$$

Its geodesic is

$$\operatorname{Geo}(H,K;u) = H^{1/2}(H^{-1/2}KH^{-1/2})^u H^{1/2}.$$

This is a complete simply connected space of nonpositive curvature; the distance, geodesic and exponential metric inequality are reviewed in [Bha13, Section 2.3]. Congruence by an invertible matrix preserves its metric whenever the congruence preserves the determinant-one locus. In particular unitary transport is an isometry. Logarithms such as $\log(H^{-1}K)$ are self-adjoint in the H metric; they need not be Hermitian in the identity frame.

The configuration space dictated by the bundle lift is

$$\mathcal{F} = \{(H_0,H_1,H_2,H_3) \in \mathcal{P}_2^4 : H_{-j} = JH_jJ^*\}. \tag{27}$$

It is the fixed set of an isometric involution of the product with metric $\sum_j A_j g_{H_j}$. It is therefore totally geodesic and geodesically convex. More explicitly it is $\mathcal{P}_2^J \times \mathcal{P}_2 \times \mathcal{P}_2^J$, with middle weight $2A_1$. Each endpoint factor consists of $\exp(ae_x)$, $a \in \mathbb{R}$, so $\dim_{\mathbb{R}} \mathcal{F} = 5$. The set is independent of θ in these frames; the energy contains the parameter.

Put

$$\Phi_{t,\theta}(H) = \frac{\pi}{L} \sum_{j \bmod 4} d_2(H_{j+1}, RH_jR^*)^2. \tag{28}$$

Every edge appears once. The relation $JR = R^{-1}J$ makes this function invariant under the involution defining (27).

Proposition 4.1. *The negative gradient flow of (28) on $\mathcal{F}$ is the restriction of the four-vertex system*

$$A_jH_j^{-1}\dot{H}_j = \frac{2\pi}{L}\Big\{\log(H_j^{-1}RH_{j-1}R^*) + \log(H_j^{-1}R^*H_{j+1}R)\Big\}. \tag{29}$$

For initial tuples in a fixed bounded set, it has a solution for every physical time $s \geq 0$, confined to a compact set independent of t and θ. Distances between two solutions with the same parameters are nonincreasing. The speed is $O(L^{-1})$, uniformly for all $s \geq 0$; the same bound holds for the time derivative of any fixed smooth tuple function on the confined range.

Proof. The negative gradient in the first variable of $\frac{1}{2}d_2(H,K)^2$ is $H\log(H^{-1}K)$. Differentiating the two incident terms and dividing by A_j gives (29). An invariant function has tangent gradient on the fixed set of an isometric involution, so (27) is preserved. The displayed logarithms are trace-free; the determinant condition is preserved.

Squared distance is jointly geodesically convex in this nonpositively curved matrix space. Hence Φ is convex, and the identity tuple I is a minimizer. The first-variation inequality for a convex function gives

$$\frac{d}{ds}\frac{1}{2}d_M(H(s),I)^2 \leq -\Phi(H(s)) \leq 0,$$

where $d_M^2 = \sum_j A_j d_2^2$ is the product squared distance. Closed bounded sets are compact in this finite-dimensional space, proving confinement and continuation for all time. Applying the same convexity inequality to the two geodesics between two solutions gives nonexpansion. On the confined range every logarithm in (29) is bounded uniformly in the unitary R. The equation then gives the speed and derivative bounds. □

The system is genuinely noncommuting. For example, choose $ab \neq 0$ and

$$H_0 = e^{a\sigma_x}, \qquad H_1 = e^{b\sigma_z}, \qquad H_2 = I, \qquad H_3 = Je^{b\sigma_z}J^* = e^{-b\sigma_z}.$$

This lies in $\mathcal{F}$. In the common frame of the first edge, H_0 and $R^*H_1R = H_1$ do not commute. Simultaneous frame conjugation cannot remove this commutator, so the construction is not restricted to independent scalar equations.

Lemma 4.2. *Set $Z_j = \log H_j$, $R_f = \frac{1}{2}\sum_j A_j \|Z_j\|^2_{\mathrm{HS}}$, and $\nu_0 = (2\pi/L)\mu_0(\theta)$. Along (29),*

$$\Phi(H) \geq \nu_0 R_f, \qquad R_f' \leq -2\Phi(H) \leq -2\nu_0 R_f. \tag{30}$$

The linearization at the identity is the negative of the effective operator with energy $(2\pi/L)q_\theta$ and mass m_A.

Proof. The logarithm based at the identity does not increase distance, so

$$d_2(H_{j+1}, RH_jR^*)^2 \geq \|Z_{j+1} - \mathrm{Ad}(R)Z_j\|^2.$$

The logarithmic tuple satisfies (22). Using (18) gives the first inequality.

Radial contraction $H_j \mapsto e^{uZ_j}$ stays in $\mathcal{F}$ and satisfies $\Phi(e^{uZ}) \leq u^2\Phi(H)$ for $0 \leq u \leq 1$. This is the distance contraction toward the common fixed point I. Differentiation from the left at $u = 1$ gives a radial derivative at least $2\Phi(H)$. Pairing the negative gradient with the gradient of R_f proves (30). Finally expansion of the logarithms in (29) at I gives the transported connection-graph operator, whose restriction was computed in Section 3. □

The parameter ν_0 includes the exact zero mode at reducibility. Thus (30) gives a uniform exponential tail in s/L on classes $\mu_0(\theta) \geq \kappa > 0$. It does not give such a tail when $\mu_0 \to 0$, although confinement and nonexpansion remain uniform.

5. Reconstruction and the actual HYM flow

On the actual rank-four bundle we use physical time s and the equation

$$h^{-1}\partial_s h = -2i\Lambda_{\omega_t}F_h. \tag{31}$$

The Einstein constant is zero. All base and bundle parameters are fixed in this equation. Relative to h_0, all metrics under consideration have determinant one. For two rank-r metrics put

$$\sigma_r(h,k) = \mathrm{tr}(h^{-1}k + k^{-1}h) - 2r. \tag{32}$$

This expression is intrinsic, nonnegative, and at least $d_r(h,k)^2$. On a fixed positive metric range it is uniformly equivalent to that squared distance. For an approximate metric k, write

$$\mathcal{R}(k) = k^{-1}\partial_s k + 2i\Lambda_{\omega_t}F_k.$$

Theorem 5.1. *For each effective trajectory of (29) with initial tuple in a fixed bounded set, and all sufficiently small t, there is a smooth determinant-one comparison metric $\widehat{K}(s)$ on $\mathcal{B}_{t,\theta}$, defined for every $s \geq 0$, with uniform positive bounds against h_0, such that*

$$\sup_{s\geq 0}\int_{X_t} |\mathcal{R}(\widehat{K}(s))|_{\mathrm{op}}\, dV_t \leq C\frac{(1+\log L)^2}{L^2}. \tag{33}$$

Let $h(s)$ solve (31) from any smooth determinant-one metric with $ch_0 \le h(0) \le Ch_0$, for fixed positive bounds. Set $\eta = \int_{X_t} \sigma_4(h(0), \widehat{K}(0))\, dV_t$ and $e_L = (1+\log L)^2/L$. Then, for every $T \ge 0$,

$$\sup_{0\le s\le TL} \int_{X_t} d_4(h(s), \widehat{K}(s))^2\, dV_t \le \eta + CTe_L. \tag{34}$$

The constants are independent of θ, t, s and T in their stated domains. The exact initial metric need not be a tensor metric.

5.1. **Equivariant reconstruction on the elliptic family.** The reconstruction uses a component correction cancelling the leading velocity, flat geodesic interpolation on the necks, and an estimate for their mismatch on the collars. We then tensor with the fixed K3 metric and descend, keeping the full contracted Chern equation. Put $\epsilon = L^{-1}$ and choose mean-zero component Green functions

$$\Delta_j g_{je} = -2\pi\delta_{p_{je}} + \frac{2\pi}{A_j}, \qquad g_{je} = \log|u| + \gamma_{je}(u), \tag{35}$$

where Δ_j is the nonnegative scalar Laplacian on D_j and u is its node coordinate. Uniqueness makes these functions equivariant under the isometry σ. The two actual incoming transports to component j are $U_{j,j-1} = R$ and $U_{j,j+1} = R^*$. Define

$$B_{je} = \log(H_j^{-1} U_{jk} H_k U_{jk}^*), \qquad \xi_j = -\sum_{e\ni j} B_{je} g_{je}, \qquad K_j = H_j e^{\epsilon\xi_j}. \tag{36}$$

The logarithms are H_j-self-adjoint and trace-free. On the punctured component, the effective equation gives the exact cancellation

$$\epsilon\Delta_j\xi_j = -H_j^{-1}\dot{H}_j. \tag{37}$$

Cut at radius $\rho = L^{-1}$. At incidence je write

$$\xi_j = -B_{je}\log|u| + c_{je} + q_{je}(u), \qquad q_{je}(0) = 0.$$

The corrected endpoint is $P_{je} = H_j e^{\epsilon c_{je}}$. On the node j,k, in the j incidence frame, use

$$N_e(u) = \mathrm{Geo}(P_{je}, U_{jk} P_{ke} U_{jk}^*; -\epsilon\log|u|). \tag{38}$$

Reversing the geodesic and using $uv = t$ gives the same metric in the other incidence frame. For each fixed time this node metric has zero Chern curvature: if P, Q are fixed endpoints, it is $P\exp(-\epsilon\log|u|\log(P^{-1}Q))$, whose logarithmic derivative has zero $\bar\partial$ away from the puncture.

Interpolate relative logarithms between K_j and N_e on $\rho < |u| < 2\rho$. Choose paired cutoffs, component transfers and node charts under the involution. On fixed components the required smooth transfer maps can be obtained from invariant lifts of base vector fields. No holomorphic product trivialization is assumed. The operations in (36)–(38) commute with unitary transport, geodesic reversal and the actual lift (11). Thus the resulting curve metric $K_E(s)$ is invariant in component *and* node frames.

We give the residual estimate for this full Chern construction. In a flat holomorphic frame, with $\omega_{E_t} = (i/2)a\, dz\wedge d\bar z$, the actual equation is

$$\partial_s H = 4a^{-1}(H_{z\bar z} - H_{\bar z} H^{-1} H_z). \tag{39}$$

In real coordinates it contains $i(H_x H^{-1} H_y - H_y H^{-1} H_x)$ as well as the two symmetric quadratic terms. This commutator is retained in all curvature estimates below.

Effective-flow confinement and speed give, uniformly for all $s \geq 0$,

$$\begin{aligned} &|B_{je}| \leq C, \qquad |\dot{B}_{je}| \leq C\epsilon, \qquad \|\xi\|_\infty \leq C(1 + |\log \rho|), \\ &\int \big(|\xi| + \epsilon^{-1}|\dot{\xi}| + |\nabla \xi|\big) \leq C, \qquad \int |\nabla \xi|^2 \leq C(1 + |\log \rho|). \end{aligned} \tag{40}$$

These integrals are over the truncated components with their fixed metrics. The identity

$$e^{-h}\partial e^h = \int_0^1 e^{-uh}(\partial h)e^{uh}\, du$$

and its differentiated form show that the component residual, after (37), is bounded by the integrals of

$$C\epsilon^2\big(|\xi| + |\xi|\,|\Delta_j \xi| + |\nabla \xi|^2\big).$$

The second derivative is contracted before taking its norm: in complex dimension one the $\bar{\partial}\partial\xi$ term in each conjugation is a scalar multiple of $\Delta_j \xi$. Differentiating the conjugating factors gives the displayed quadratic-gradient terms. Since $\Delta_j \xi$ on the punctured component is bounded, the residual is $O(\epsilon^2(1 + |\log \rho|))$ after integration. The time derivative of the correction has the same bound by (40) and $H_j^{-1}\dot{H}_j = O(\epsilon)$.

Here is the collar estimate which allows matching without losing this order. For small endpoint logarithms C_j, C_k, let $D(C_j, C_k; u)$ be the relative logarithm at H_j of $\mathrm{Geo}(H_j e^{C_j}, U_{jk} H_k e^{C_k} U_{jk}^*; u)$, minus $C_j + uB_{je}$. On the confined matrix range it is smooth, and

$$D(0, 0; u) = 0, \qquad D(C_j, C_k; 0) = 0.$$

Bounded mixed derivatives imply $|D| \leq C(|C_j| + |C_k|)|u|$ and $|\partial_u D| + |\partial_u^2 D| \leq C(|C_j| + |C_k|)$. Substitute $C_j = \epsilon c_{je}$, $C_k = \epsilon c_{ke}$ and $u = -\epsilon \log |z|$. Including the regular part q_{je} gives the mismatch bound, through two spatial derivatives,

$$C\big(\epsilon\rho^{1-k} + \epsilon^2(1 + |\log \rho|)\rho^{-k}\big), \qquad k = 0, 1, 2. \tag{41}$$

In the second derivative the additional term of order $\epsilon^3\rho^{-2}$ is absorbed by the displayed bound.

The curvature difference from the exactly flat node profile contains the second derivative of this mismatch, its first derivative times the node logarithmic derivative $O(\epsilon/\rho)$, and its zeroth derivative times the corresponding second-derivative and quadratic terms. Cutoff derivatives have the same orders in ρ. Since the collar area is $O(\rho^2)$, its integrated residual is

$$C\big(\epsilon\rho + \epsilon^2(1 + |\log \rho|)^2\big).$$

The middle-node time residual is $O(\epsilon\rho^2)$; its spatial Chern curvature is zero. Transfer to the actual smooth fibre adds $C|t|\rho^{-N}$ for a fixed finite N, by the smooth plumbing metric coefficients and the two derivatives in the contracted equation. Putting $\rho = L^{-1}$ proves

$$\sup_{s \geq 0} \int_{E_t} |K_E^{-1}\dot{K}_E + 2i\Lambda_{E_t} F_{K_E}|_{\mathrm{op}}\, dA \leq C\frac{(1 + \log L)^2}{L^2}. \tag{42}$$

Every time-dependent input was a confined tuple or its $O(L^{-1})$ velocity. Thus this is a residual bound on the whole physical half-line, including reducible parameters.

5.2. **Exact tensor curvature and descent.** On the cover set $\widetilde{K} = h_S \otimes K_E$. Product Chern connections give exact identities

$$\begin{aligned} D_{\widetilde{K}} &= D_S \otimes I + I \otimes D_{K_E}, \\ F_{\widetilde{K}} &= F_{T^{1,0}S} \otimes I + I \otimes F_{K_E}, \\ \mathcal{R}(\widetilde{K}) &= I_{T^{1,0}S} \otimes (K_E^{-1}\dot{K}_E + 2i\Lambda_{E_t}F_{K_E}). \end{aligned} \tag{43}$$

The last equality uses the zero mean curvature of h_S. Equivariance proved above makes $\widetilde{K}$ descend to $\widehat{K}$. Its determinant is one in the fixed trivialization. The operator norm of the tensor residual is unchanged, whereas integration divides by the degree two of the cover. Consequently (42) implies (33).

For two tensor metrics there is also the exact distance identity

$$d_4(h_S \otimes H, h_S \otimes K)^2 = 2d_2(H, K)^2. \tag{44}$$

Together with (13), division by the cover degree makes their global quotient squared distance equal to the curve integral. Neither tensor identity is assumed for an arbitrary exact solution $h(s)$.

5.3. **Intrinsic comparison for arbitrary rank-four metrics.** On each fixed compact smooth fibre, the classical HYM metric flow exists for all time; see [HKKP18, Section 4.1, after (4.8)]. Order comparison [HKKP18, Proposition 4.3] with ch_0 and Ch_0 preserves the bounds, and the scalar determinant equation preserves the normalization. These fixed-fibre statements impose no uniform Sobolev-constant hypothesis on the degenerating family.

For an exact flow h and a comparison metric k in a common fixed positive range, the Chern trace calculation gives

$$(\partial_s + \Delta_{\omega_t})\sigma_4(h, k) \le C|\mathcal{R}(k)|_{\mathrm{op}}. \tag{45}$$

For two exact velocities, the intrinsic trace calculation of [Wan08, Proposition 2.4], with the coupling section set to zero, makes the left side a sum of nonpositive quadratic derivative terms in local holomorphic frames. Replacing the velocity of k by its exact-flow velocity plus $k\mathcal{R}(k)$ adds the traces of $h^{-1}k\mathcal{R}(k)$ and its inverse-metric counterpart. Their absolute values are bounded by the right side under the fixed relative bounds. The same calculation is summed over all complex coordinate directions. It is intrinsic and therefore applies across the nonflat tangent-bundle frames, without reducing the exact solution to a tensor ansatz.

Proof of Theorem 5.1. The equivariant construction and (43) prove the comparison metric and its residual estimate. Integrate (45) on the closed smooth fibre, so that the Laplacian integrates to zero. Its accumulated right side up to time TL is at most $CT(1 + \log L)^2/L$. Since $d_4^2 \le \sigma_4$, this proves (34). No derivative bound on the exact initial metric enters this argument. □

Theorem 5.2. *Under the hypotheses of Theorem 5.1, assume also $\mu_0(\theta) \ge \kappa > 0$. For sufficiently small t,*

$$\sup_{s \ge 0} \int_{X_t} d_4(h(s), \widehat{K}(s))^2 \, dV_t \le C\eta + C_\kappa e_L(1 + |\log e_L|). \tag{46}$$

Proof. For the actual flow write $h = h_0 H = h_0 e^{\Psi}$. Pointwise $\operatorname{tr}\Psi = 0$. The exact relative Chern equation is

$$H_s = -D_\theta^* D_\theta H - 4\sum_\alpha (D_\alpha H)^* H^{-1} (D_\alpha H).$$

The contracted background-curvature commutator is zero. All quadratic terms are retained, including those from noncommuting derivatives. Put $F(s) = \int_{X_t} \operatorname{tr}(e^{\Psi} - I)\, dV_t$. On the fixed positive range, Taylor's formula and trace freeness give

$$\frac{c}{2}\|\Psi\|_{L^2}^2 \le F \le \frac{C}{2}\|\Psi\|_{L^2}^2, \qquad -F' \ge \frac{c^2}{C}\int_{X_t} |D_\theta \Psi|^2\, dV_t.$$

For the derivative inequality first bound $H^{-1} \ge C^{-1} I$ inside the positive quadratic term. The unweighted trace then satisfies $4\sum_\alpha \operatorname{tr}((D_\alpha H)^* D_\alpha H) = |D_\theta H|^2$. The divided differences of the exponential on the bounded spectrum are at least c, giving $|D_\theta H| \ge c|D_\theta \Psi|$. No commutator is discarded inside a matrix-weighted trace.

There is no trace-free parallel kernel on the stated parameter class. Theorem 3.1 therefore gives

$$F(s) \le F(0) \exp\left(-\frac{2c^2}{C^2}(1 - \varepsilon(t))\frac{2\pi\mu_0(\theta)}{L} s\right).$$

Lemma 4.2 gives a corresponding exponential tail for the effective trajectory. The reconstruction is uniformly Lipschitz in global L^2 on the confined tuple range: the Green coefficients have bounded L^2 norms multiplied by L^{-1}, and the node and collar maps have bounded tuple derivatives. It sends the identity tuple exactly to h_0. Equation (44) therefore transfers effective-flow decay to the comparison metric.

For $s \ge TL$, the exact and reconstructed squared distances to h_0 are at most $Ce^{-c_\kappa T}$. For $s \le TL$, use (34). Taking T proportional to $1 + |\log e_L|$ and applying the squared triangle inequality proves (46). □

The all-time conclusion is uniform on a nonempty stable parameter class. At reducibility there are nonidentity parallel metrics, and as $\mu_0 \to 0$ the above tail constants degenerate. The bounded-time comparison remains uniform there; extending it to the weakest inverse rate requires a further argument.

6. Phase selection on a fixed Hermitian–Einstein bundle

The component metrics required for spatial initial data are selected by an intrinsic fixed-background argument. Let (Y, ω) be a connected compact Kähler manifold of volume V. Let (E, h_0) be a Hermitian–Einstein bundle of rank r, with Chern connection D and $i\Lambda F_D = \lambda I$. Write $h = h_0 H$ and use the normalized equation

$$H^{-1} H_s = -2(i\Lambda F_h - \lambda I), \qquad \det H = 1. \tag{47}$$

The base and connection are fixed throughout this section. The positive gap used here is not asserted to be uniform on a degenerating family.

Let $\mathcal{A}$ be the algebra of D-parallel complex endomorphisms, and let Π be the L^2 orthogonal projection onto $\mathcal{A}$. Denote by $\lambda_\perp > 0$ the first positive eigenvalue of D^*D on self-adjoint endomorphisms, with its parallel kernel omitted.

Theorem 6.1. *Suppose $H(0)$ is smooth, $\det H(0) = 1$ and $cI \le H(0) \le CI$, where $0 < c \le C < \infty$. There is a unique positive parallel endomorphism $P = P(H(0))$, with $\det P = 1$, to which $H(s)$ converges in L^2 and uniformly. Put*

$$M = \Pi H, \qquad \mathcal{J} = \log\det M, \qquad W = V^{-1}\int_Y |H - M|^2\, dV, \qquad \gamma = \frac{2c^2\lambda_\perp}{C^2}. \tag{48}$$

Then M decreases in matrix order to P, and

$$\frac{W}{2C^2} \le \mathcal{J} \le \frac{W}{2c^2}, \qquad \mathcal{J}(s) \le \mathcal{J}(0)e^{-\gamma s}, \qquad \operatorname{tr}(M(s)-P) \le C\mathcal{J}(s). \tag{49}$$

In particular,

$$V^{-1}\int_Y |H(s)-P|^2\,dV \le C^2(2\mathcal{J}(0)+\mathcal{J}(0)^2)e^{-\gamma s}. \tag{50}$$

For $s \ge 1$ the squared affine distance satisfies

$$\sup_Y d_r(H(s),P)^2 \le K_Y(2\mathcal{J}(0)+\mathcal{J}(0)^2)e^{-\gamma(s-1)}. \tag{51}$$

Here K_Y also depends on c, C, r and the fixed background. For two initial metrics in a common positive range, the selected phases obey

$$V\sigma_r(P(H(0)),P(K(0))) \le \int_Y \sigma_r(H(0),K(0))\,dV. \tag{52}$$

Consequently phase selection is Lipschitz on such a range. If $\mathcal{A} = \mathbb{C}I$, the selected phase is I.

Proof. Decompose the unitary holonomy representation at a base point into inequivalent irreducible representations and their multiplicity spaces. Parallel transport of the invariant summands gives a global orthogonal parallel decomposition

$$E = \bigoplus_a F_a \otimes \mathbb{C}^{m_a}, \qquad D = \bigoplus_a D_a \otimes I, \qquad \mathcal{A} = \bigoplus_a I_{F_a} \otimes \operatorname{End}(\mathbb{C}^{m_a}). \tag{53}$$

The transported subspaces are independent of the chosen path because they are holonomy invariant. This does not require individual parallel vectors on F_a. The splitting is also holomorphic since the $(0,1)$ part of D is the holomorphic structure.

If $r_a = \operatorname{rank} F_a$ and B_{aa} is a diagonal block of an endomorphism section, direct testing against every parallel endomorphism gives

$$\Pi B = \bigoplus_a I_{F_a} \otimes \frac{1}{r_a V}\int_Y \operatorname{tr}_{F_a} B_{aa}\,dV. \tag{54}$$

The partial trace is globally defined using the fixed metric on F_a. Diagonal restriction, partial trace and integration preserve positivity. Thus Π is positive, unital and adjoint preserving. It preserves the normalized integrated trace, and, for any parallel endomorphism B,

$$\operatorname{tr}(B\Pi Q) = V^{-1}\int_Y \operatorname{tr}(BQ)\,dV. \tag{55}$$

Order comparison for (47) preserves $cI \le H \le CI$; its scalar determinant equation preserves $\det H = 1$. Hence $cI \le M \le CI$.

We need the full relative metric equation. In unitary normal base coordinates with $\omega = (i/2)\sum dz_\alpha \wedge d\bar{z}_\alpha$ at the point, expansion of $\bar{\partial}_D(H^{-1}\partial_D H)$ gives

$$H_s = -D^*DH - Q(H), \qquad Q(H) = 4\sum_\alpha (D_\alpha H)^* H^{-1}(D_\alpha H) \succeq 0. \tag{56}$$

The second-order term has scalar principal part $4\sum \partial_\alpha \partial_{\bar{\alpha}} H$. Its difference from $-D^*DH$ on endomorphism-valued zero-forms is the contracted curvature commutator, which vanishes because $i\Lambda F_D = \lambda I$. Differentiating H^{-1} gives precisely the displayed quadratic term. In

particular, it cannot be dropped for matrix-valued H. Since $\Pi D^*DH = 0$, positivity of the projection proves

$$M' = -\Pi Q(H) \preceq 0. \tag{57}$$

It follows that M has a parallel positive limit P between cI and CI.

Taylor's integral remainder for $\log\det$ on the segment from M to H has Hessian between $-c^{-2}$ and $-C^{-2}$ in Frobenius norm. The integrated linear term vanishes, since

$$V^{-1}\int_Y \operatorname{tr}(M^{-1}(H-M))\,dV = 0$$

by (55). Integrating $\log\det H = 0$ gives the two bounds for $\mathcal{J}$ in (49). Next, the same trace identity and positivity give

$$\begin{aligned} -\mathcal{J}' &= V^{-1}\int_Y \operatorname{tr}(M^{-1}Q(H))\,dV \\ &\geq C^{-2}V^{-1}\int_Y |DH|^2\,dV \geq C^{-2}\lambda_\perp W \geq \gamma\mathcal{J}. \end{aligned}$$

For the first inequality, remove both positive matrix weights using their lower bounds. Only then use $4\sum_\alpha \operatorname{tr}((D_\alpha H)^* D_\alpha H) = |DH|^2$. The gap applies because $H - M$ is orthogonal to the whole parallel algebra and $D(H-M) = DH$. This proves exponential decay of $\mathcal{J}$.

By continuity, $\log\det P = 0$. Integrating the derivative of $\log\det$ on the positive segment from P to M gives $\operatorname{tr}(M-P) \leq C\mathcal{J}$. Orthogonality to parallel sections gives the exact decomposition

$$V^{-1}\int_Y |H-P|^2\,dV = W + |M-P|^2 \leq 2C^2\mathcal{J} + C^2\mathcal{J}^2.$$

Here $M - P \succeq 0$, so its norm is at most its trace. This proves (50).

The metric $h_0 P$ has the same Chern connection as h_0 and is stationary. The unforced trace comparison from (45), followed by the time-one scalar heat kernel on the fixed compact base, bounds the supremum of $\sigma_r(H(s), P)$ by a fixed multiple of its integral at $s-1$. Equivalence of trace distance and squared norm on the fixed range proves (51). This uses no derivative bound on arbitrary bounded initial metrics and no degenerating heat kernel.

Finally, integrate the trace comparison for two exact flows. Uniform convergence to their parallel limits proves (52). It also gives uniqueness, and the equivalence of the distances gives the stated Lipschitz dependence. A positive scalar endomorphism with determinant one is the identity. □

On a bundle with repeated irreducible factors the phase belongs to the actual algebra (53); it is not a constant matrix in a chosen local frame on the whole bundle. In the next section the fixed component bundles supply this algebra and its positive gap. The global degeneration estimate will be established separately.

7. Entry from spatial initial metrics

The four fixed component problems on the cover have bases and bundles

$$Y_j = S \times D_j, \qquad F_j = p_S^* T^{1,0}S \otimes \mathbb{C}^2, \qquad h_{j,0} = h_S \otimes I. \tag{58}$$

They have zero Einstein constant and complex parallel algebra $I_{T^{1,0}S} \otimes \operatorname{End}(\mathbb{C}^2)$. Their geometry is independent of t and θ; the latter enters the node identifications. Thus Theorem 6.1 selects, from each smooth determinant-one component metric, a phase $I_{T^{1,0}S} \otimes P_j$. Positivity and $(\det P_j)^2 = 1$ give $\det P_j = 1$. For equivariant initial profiles, uniqueness of

the component flow gives $P_{-j} = JP_jJ^*$. The selected tuple belongs to the finite-dimensional configuration space of Section 4.

7.1. **The open class and its topology.** Put $\rho = L^{-1}$ and let $\Omega_j(2\rho) \subset D_j$ be the complement of the two open node discs of radius 2ρ. Use the equivariant smooth component transfers already fixed in the reconstruction. An actual initial metric on $\mathcal{B}_{t,\theta}$ pulls back to $h_0 e^{\Psi_{\mathrm{in}}}$, where Ψ_{in} is a smooth equivariant trace-free h_0-Hermitian endomorphism. Its restriction to the jth component exterior is a section Ψ_j on $S \times \Omega_j(2\rho)$ with values in $\mathrm{Herm}_0(F_j)$.

Fix $0 < \beta < 1$ and $R_0 > 0$. Using the fixed component connections and metrics in all component norms, define

$$\mathcal{N}_t(\Psi) = \|\Psi\|_{C^0(X_t)} + \inf\left\{ \sum_{j=0}^{3} \|\widetilde{\Psi}_j\|_{C^{6,\beta}(Y_j)} : \widetilde{\Psi}_j|_{S\times\Omega_j(2\rho)} = \Psi_j, \right. \\ \left. (\widetilde{\Psi}_j)_{j=0}^3 \text{ is an equivariant smooth extension} \right\}, \tag{59}$$

$$\mathcal{B}_{R_0}(t,\theta) = \{h_0 e^{\Psi} : \mathcal{N}_t(\Psi) < R_0\}.$$

Equivariance includes the base involution and the actual component bundle lift $(i\,d\iota) \otimes J$, not just conjugation of a matrix.

Lemma 7.1. *For every fixed smooth fibre, $\mathcal{B}_{R_0}(t,\theta)$ is a nonempty open set in the $C^{6,\beta}$ topology on smooth determinant-one metrics. It contains a parameter-independent small ball of smooth equivariant component profiles, joined across the nodes, and contains metrics which are not tensor products.*

Proof. Each component exterior is a compact manifold with smooth boundary. Smooth collar extension operators for its bundle sections are bounded in the fixed $C^{6,\beta}$ norms. Averaging the operators over the finite involution preserves equivariance, Hermitian symmetry and zero trace. For a fixed fibre their operator bounds, together with the triangle inequality for the infimum in (59), prove continuity of $\mathcal{N}_t$. The global C^0 term makes it a norm, and exponentiation is smooth. No uniform extension-operator bound as $\rho \to 0$ is used. In particular the radius in (59) is not a radius for an unrestricted global C^0 norm.

Take a sufficiently small fixed ball of equivariant smooth component logarithms and join their metrics by fibrewise geodesics on the necks. The original profiles are admissible extensions; compactness of the positive matrix range bounds the interpolated logarithms. This yields members of the asserted class for all small t. Smooth modifications supported in the middle necks with sufficiently small C^0 amplitude are allowed, independently of their derivatives.

For a non-tensor example, take a nonzero local section $A(x) \in \mathrm{Herm}_0(T^{1,0}S)$ and a real smooth function $\chi(z)$ supported away from the nodes. Choose the support disjoint from its deck image, possible because ι is free, and average the transported logarithm $\chi(z)A(x) \otimes \sigma_z$ under the deck action. Its mixed trace-free component remains nonzero on the original support. Small multiples belong to the class. For positive G, K,

$$\log(G \otimes K) = (\log G) \otimes I + I \otimes (\log K),$$

whose projection to $\mathrm{Herm}_0(T^{1,0}S) \otimes \mathrm{Herm}_0(\mathbb{C}^2)$ is zero. Uniqueness of the Hermitian logarithm therefore excludes tensor factorization of the example. These initial metrics can depend on all six real spatial coordinates and need not commute with the tangent curvature. □

7.2. **Component evolution and the choice of phase.** Choose an equivariant extension tuple in (59) with total norm less than $2R_0$. Let $\Phi_j(s)$ be its relative metric under the exact HYM flow on Y_j. Theorem 6.1 and the fixed component gaps give constants depending only on R_0 and the fixed geometry such that

$$\sup_{Y_j} d_4(\Phi_j(s), I \otimes P_j)^2 \le C_{R_0} e^{-\gamma_{R_0}(s-1)}, \qquad s \ge 1. \tag{60}$$

All these flows remain in a common compact positive range. For every finite integer n, there is a common constant C_n bounding their spatial covariant derivatives through order three and their time derivatives on $0 \le s \le n$. Indeed, the bounded $C^{6,\beta}$ initial set is precompact in $C^{4,\beta/2}$ on the fixed compact Y_j. The relative equation has scalar elliptic principal part and a smooth quadratic-gradient nonlinearity on the fixed positive range. Local parabolic continuous dependence, smoothing and fixed-base global continuation give bounds on a neighbourhood of each initial datum in the compact closure. A finite cover gives C_n. Enlarge them to be nondecreasing and at least one. This argument uses component derivative control, rather than drawing derivative estimates from an arbitrary C^0 bound.

Two allowed extensions of the same initial metric differ only on S times discs of total area $O(\rho^2)$ and remain in a fixed positive range. Their initial trace-distance integral is therefore $O_{R_0}(\rho^2)$. The phase contraction (52) implies

$$d_2(P_j, P_j')^2 \le C_{R_0}\rho^2. \tag{61}$$

The factor two in the rank-four tensor distance and the fixed component volume are included in the constant. Thus the phase is determined up to $O(\rho)$, without claiming an exactly canonical extension at finite t. Effective-flow nonexpansion preserves this bound at all times.

Theorem 7.2. *There is a preparation time $s_*(t) \to \infty$, with $s_*(t)/L \to 0$, independent of θ and of the initial metric in $\mathcal{B}_{R_0}(t,\theta)$, with the following property. Let P be a component phase tuple selected as above. Let $H_f(s)$ solve the effective equation with $H_f(0) = P$, and let $\widehat{K}(H_f(s))$ be its reconstruction. The actual HYM solution from any metric in this open class satisfies*

$$\eta_* := \int_{X_t} \sigma_4(h(s_*), \widehat{K}(P))\, dV_t \le C_{R_0}\left(\sqrt{\delta_L} + e^{-\gamma_{R_0}(s_*-1)} + L^{-2}\right) \longrightarrow 0, \tag{62}$$
$$\delta_L = \frac{1+\log L}{L} + |t| L^N,$$

where N is a fixed finite exponent. For each $T \ge 0$,

$$\sup_{s_* \le s \le s_*+TL} \int_{X_t} d_4(h(s), \widehat{K}(H_f(s-s_*)))^2\, dV_t \le \eta_* + C_{R_0} T e_L. \tag{63}$$

If also $\mu_0(\theta) \ge \kappa > 0$, then

$$\sup_{s \ge s_*} \int_{X_t} d_4(h(s), \widehat{K}(H_f(s)))^2\, dV_t \le C_{R_0}\eta_* + C_{R_0,\kappa} e_L(1 + |\log e_L|) + C_{R_0}(s_*/L)^2. \tag{64}$$

All estimates before the last one are uniform through $\theta \in \pi\mathbb{Z}$.

Proof. On the component exteriors transfer $\Phi_j(s,x,z)$ to the actual smooth fibre. At incidence je put $A_{je}(s,x) = \Phi_j(s,x,p_{je})$. In a common node frame join the two transported endpoints by the rank-four affine geodesic with parameter $-L^{-1}\log|u|$. At the oriented node $j, j+1$, the right endpoint is

$$(I \otimes R^*) A_{j+1,e}(I \otimes R),$$

as follows from the actual incidence relation $w_{j+1} = Rz_j$. Interpolate to the component metrics on $\rho < |u| < 2\rho$ by equivariant relative-logarithm cutoffs. All these constructions are intrinsic in the positive metrics on $T^{1,0}S \otimes \mathbb{C}^2$ and respect the deck lift. They define a descended preparation metric $K_{\mathrm{pre}}(s)$ in a fixed positive range. At $s = 0$ it agrees with the actual initial metric on the component exteriors.

We estimate its full contracted Chern residual for $0 \le s \le n$. On component regions the comparison solves the full rank-four equation for the fixed product metric, including tangent curvature and all spatially varying noncommuting terms. The only transfer error is from the elliptic geometry. Smooth plumbing coefficients, with the required two derivatives and the C_n bounds, give integrated error $C_n|t|\rho^{-N}$ for a fixed finite N independent of n.

On a node interior, for each fixed $x \in S$ the geodesic has zero Chern curvature in the curve direction. Its S-direction curvature generally does not vanish. Covariant S derivatives through order two of the geodesic are bounded by C_n on the compact endpoint range. The tangent background has zero mean curvature, so the S-contracted residual is $O(C_n)$. The time residual has the same bound. Mixed S/curve curvature components can be nonzero but contract to zero with the exact block diagonal product Kähler form. Since the middle-node volume is $O(\rho^2)$, these terms contribute $C_n\rho^2$ in integral. Thus the argument does not assert vanishing of the full spatial curvature.

On an incidence collar the component endpoint discrepancy and the geodesic discrepancy, with their first two curve derivatives, obey

$$C_n\big(\rho + L^{-1}|\log\rho|\big)\rho^{-k}, \qquad k = 0, 1, 2. \tag{65}$$

For the component term this is Taylor expansion in the curve coordinate; for the geodesic term it is differentiation of the smooth endpoint map at parameter $O(L^{-1}|\log\rho|)$. The relevant mixed derivatives are included in C_n. Expanding the contracted Chern expression relative to the node profile, the second derivatives of the discrepancy and the cutoff terms integrate to $C_n(\rho + L^{-1}|\log\rho|)$. Products with the node derivative $O((L\rho)^{-1})$ satisfy the same bound, as do quadratic mismatch terms for small $\rho + L^{-1}|\log\rho|$. This calculation is valid for arbitrary rank-four matrices. The cutoff is independent of S, so S derivatives and time derivatives still contribute only $C_n\rho^2$. Mixed contractions remain zero. Taking $\rho = L^{-1}$ therefore proves

$$\sup_{0\le s\le n} \int_{X_t} |\mathcal{R}(K_{\mathrm{pre}}(s))|_{\mathrm{op}}\, dV_t \le C_n\delta_L. \tag{66}$$

At time zero the exact and preparation metrics differ only in regions of volume $O(\rho^2)$. Boundedness gives initial integrated trace distance at most $C_{R_0}\rho^2$, regardless of the initial neck derivatives. The forced trace comparison gives

$$\sup_{0\le s\le n} \int_{X_t} d_4(h(s), K_{\mathrm{pre}}(s))^2\, dV_t \le C_{R_0}\rho^2 + nC_n\delta_L. \tag{67}$$

Choose

$$s_* = \max\{n \in \mathbb{N} : nC_n \le \delta_L^{-1/2}\}. \tag{68}$$

For small t this set is nonempty and finite. Each fixed integer eventually belongs to it, so $s_* \to \infty$. Moreover $s_*C_{s_*}\delta_L \le \sqrt{\delta_L}$ and $s_*/L \le \sqrt{\delta_L}$, using $C_n \ge 1$ and $\delta_L \ge L^{-1}$. This is an implicit common time; no logarithmic formula for it is claimed.

On the component exteriors at s_*, use (60). The Green correction in $\widehat{K}(P)$ has global L^2 size $O(L^{-1})$. On the remaining node regions boundedness and volume $O(\rho^2)$ suffice.

Therefore

$$\int_{X_t} d_4(K_{\mathrm{pre}}(s_*), \widehat{K}(P))^2 \, dV_t \le C_{R_0}(e^{-\gamma_{R_0}(s_*-1)} + L^{-2}).$$

Combine this with (67) and the equivalence of trace distance and squared affine distance on the fixed positive range to obtain (62). Apply Theorem 5.1 starting at s_* to obtain (63).

The effective speed is $O(L^{-1})$ for every time, so $d_M(H_f(s-s_*), H_f(s)) \le C_{R_0} s_*/L$ for $s \ge s_*$, where d_M is the mass-weighted product distance. The reconstruction is uniformly Lipschitz into global L^2 on the confined tuple range. Thus changing to the physical-time-aligned comparison costs at most $C_{R_0}(s_*/L)^2$, with fixed squared-triangle constants. Under the positive-rate hypothesis, Theorem 5.2 applied after entry proves (64). □

The extension ambiguity (61), effective-flow nonexpansion and the reconstruction bound show that changing the allowed component extensions changes all subsequent comparisons by $O(L^{-1})$ in L^2 distance. The theorem gives entry from a fixed-radius open class with genuine spatial variation. It does not give trajectory tracking during the preparation interval, uniform derivative estimates over the whole neck, or an all-time bound as μ_0 tends to zero.

8. Transport recovery and the order of limits

8.1. **The observed invariant.** In this inverse problem the compact family and its unitary one-parameter bundle class are known. The normalization $L = \log(1/|t|)$ and the three component areas A_0, A_1, A_2 are known. The parameter θ, the transition matrices and their cycle product are not supplied. The observations are the five lowest eigenvalues of the actual operator $\mathcal{L}_{t,\theta}$, including its zero modes. The target is the conjugacy class of $\mathrm{Ad}(U_\theta)$ in $\mathrm{SO}(3)$. It is a global bundle-transport invariant. The integral Chern number computed in Section 2 is not an output of this spectral inverse.

Theorem 8.1. *With the above observations define*

$$\widehat{s}_t = \frac{A_0 A_1^3 A_2}{8} \left(\frac{L}{2\pi}\right)^5 \prod_{k=0}^{4} \lambda_k(t,\theta). \tag{69}$$

For the error in Theorem 3.1,

$$(1-\varepsilon(t))^5 \sin^2\theta \le \widehat{s}_t \le (1+\varepsilon(t))^5 \sin^2\theta. \tag{70}$$

Thus $3-4\widehat{s}_t$ *recovers* $\operatorname{tr}\mathrm{Ad}(U_\theta)$ *with error* $O(\varepsilon(t)\sin^2\theta)$. *If* $\mathrm{clip}(x) = \max(0, \min(1,x))$, *then*

$$\widehat{\alpha}_t = 2\arcsin\sqrt{\mathrm{clip}(\widehat{s}_t)} \tag{71}$$

converges uniformly over the family to its principal projective conjugacy angle in $[0,\pi]$. *On a class* $\sin^2\theta \in [\kappa, 1-\kappa]$, $0 < \kappa < 1/2$, *the angle error is at most* $C_{A,\kappa}(\varepsilon(t)+\delta)$ *if each observed rescaled eigenvalue* $L\lambda_k/(2\pi)$ *has additive error at most* δ, *for small* δ *and* t.

Proof. In coordinates (a,x,y,z,b) write $c_\theta = \cos\theta$ and $s_\theta = \sin\theta$. The energy and mass matrices are

$$Q_\theta = \begin{pmatrix} 2 & -2 & 0 & 0 & 0 \\ -2 & 4 & 0 & 0 & -2c_\theta \\ 0 & 0 & 4 & 0 & -2s_\theta \\ 0 & 0 & 0 & 4 & 0 \\ 0 & -2c_\theta & -2s_\theta & 0 & 2 \end{pmatrix}, \qquad M_A = \mathrm{diag}(A_0, 2A_1, 2A_1, 2A_1, A_2). \tag{72}$$

The Schur complement of the middle $4I_3$ block is $\left(\begin{smallmatrix} 1 & -c_\theta \\ -c_\theta & 1 \end{smallmatrix}\right)$. Consequently

$$\det Q_\theta = 64\sin^2\theta, \qquad \prod_{k=0}^{4}\mu_k(\theta) = \frac{8\sin^2\theta}{A_0A_1^3A_2}. \tag{73}$$

Multiplying the five relative bounds in (19) proves (70). At $\theta \in \pi\mathbb{Z}$ both products are exactly zero.

Since $U_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta})$, its adjoint rotation has trace $1 + 2\cos(2\theta) = 3 - 4\sin^2\theta$. The principal angle is $2\arcsin\sqrt{\sin^2\theta}$. Uniform continuity on $[0,1]$ proves uniform recovery after clipping. On the stated interior class the derivative of this function is bounded, which proves Lipschitz dependence on the estimator.

The rescaled eigenvalues lie in a fixed bounded interval by (19). Their product is a polynomial with bounded derivative on a fixed bounded enlargement. Errors at most δ in the five factors therefore change (69) by at most $C_A\delta$. Combine this with the preceding angle bound. □

No uniform Lipschitz angle bound is asserted at the endpoint angles. In particular, fixed absolute noise need not permit relative recovery of an arbitrarily small angle. The observations do not distinguish a signed SU(2) lift, central scalar twists, or reversal of the cycle orientation. Nor does this one-parameter statement imply spectral rigidity among arbitrary bundles outside the specified family.

Corollary 8.2. *Let ϑ be the small signed distance from θ to $\pi\mathbb{Z}$. Jointly as $t \to 0$ and $\vartheta \to 0$ with $\vartheta \neq 0$,*

$$\lambda_0(t,\theta) = \frac{2\pi\vartheta^2}{L(A_0 + 2A_1 + A_2)}(1 + o(1)). \tag{74}$$

At $\vartheta = 0$ the eigenvalue is exactly zero.

Proof. At $\theta = 0$ the kernel is spanned by $w = (1,1,0,0,1)^{\mathsf{T}}$. Direct cofactors give $\operatorname{adj}(Q_0) = 64ww^{\mathsf{T}}$. The coefficient of $-\mu$ in $\det(Q_0 - \mu M_A)/\det M_A$, equal to the product of its four positive generalized eigenvalues, is therefore

$$\frac{\operatorname{tr}(\operatorname{adj}(Q_0)M_A)}{\det M_A} = \frac{8(A_0 + 2A_1 + A_2)}{A_0A_1^3A_2}.$$

The simple small eigenvalue is analytic and even in θ: reflection of the y coordinate identifies the matrices at θ and $-\theta$. Divide (73) by the analytic nonzero product of the other four eigenvalues to get

$$\mu_0(\theta) = \frac{\theta^2}{A_0 + 2A_1 + A_2} + O(\theta^4).$$

A sign change of the b coordinate reduces the calculation at any other multiple of π to this one. The relative, parameter-uniform bound in Theorem 3.1 proves the joint assertion. □

The symbolic determinant and cofactor identities in this proof are reproduced by the accompanying computation. They check the finite matrix normalization; they do not replace the proof relating its eigenvalues to the actual threefold operator.

8.2. **A common space for the time limits.** To compare metrics on different fibres, pull them to the cover and use the fixed component frames on compact subsets of $S \times D_j$ away from the nodes. All limits below mean L^2 convergence on every such compact subset, relative to $h_S \otimes I$. An effective tuple P denotes the corresponding relative metrics $I_{T^{1,0}S} \otimes P_j$. The reconstruction differs from its constant tuple by $O(L^{-1})$ on these subsets. Global L^2 comparison implies this component convergence. Write $H_\theta(\tau; P)$ for the effective flow in rescaled time $\tau = s/L$, initialized at P. Its equation is independent of t.

Proposition 8.3. *For matching initial data $h_{t,\theta}(0) = \widehat{K}_{t,\theta}(P)$ the following hold.*

(1) *For fixed $\theta \notin \pi\mathbb{Z}$ and an invariant tuple $P \neq I$, physical time and degeneration do not commute:*

$$\lim_{s\to\infty} \lim_{t\to 0} h_{t,\theta}(s) = P, \qquad \lim_{t\to 0} \lim_{s\to\infty} h_{t,\theta}(s) = I. \tag{75}$$

(2) *On a fixed rate class $\mu_0(\theta) \geq \kappa > 0$, for fixed θ and P, the limit at each $\tau \geq 0$ is $H_\theta(\tau; P)$. The limits $t \to 0$ and $\tau \to \infty$ commute and equal I. More generally, for bounded tuples and every sequence $t \to 0$, $\tau(t) \to \infty$, with θ in this rate class, $h_{t,\theta}(L\tau(t)) \to I$.*

(3) *If $\theta(t) \to 0$ with $\theta(t) \notin \pi\mathbb{Z}$, take $P_j = \exp(a\sigma_x)$ for all j, with $a \neq 0$. Then*

$$\lim_{\tau\to\infty} \lim_{t\to 0} h_{t,\theta(t)}(L\tau) = P, \qquad \lim_{t\to 0} \lim_{\tau\to\infty} h_{t,\theta(t)}(L\tau) = I. \tag{76}$$

Proof. For a fixed physical time s, effective speed $O(L^{-1})$ makes the effective tuple tend to P. In (34) take $\eta = 0$ and $T = s/L$. Its constant is independent of T, so the error is at most $Cs(1 + \log L)^2/L^2$, which tends to zero. This proves the first limit in (75). On each fixed stable smooth fibre, the relative logarithm has no parallel trace-free kernel. The coercivity argument in the proof of Theorem 5.2, using the fixed-fibre positive gap, implies convergence to h_0. This proves the other limit.

On the rate class in part 2, the all-time comparison has error tending to zero uniformly, and the effective trajectory tends to I exponentially in τ with uniform constants on bounded tuples. These facts prove both the fixed-τ limit and the uniform diagonal statement. They also prove the claimed interchange of the two limits.

At $\theta = 0$ the tuple in part 3 is stationary: $R = I$ and all its entries coincide. It is invariant under J and is a global parallel metric at that parameter. On each bounded τ interval, smooth dependence of the confined effective vector fields on θ gives $H_{\theta(t)}(\tau; P) \to P$ uniformly. The bounded-time actual-flow estimate is uniform in θ, hence the actual metrics have the same limit. Each fixed nonzero t, on the other hand, has a stable parameter and converges to I at infinite time. This proves (76). □

Corollary 8.4. *For the entire bounded open class of Theorem 7.2, uniform diagonal relaxation to I holds when $\mu_0(\theta_t) \geq \kappa > 0$, $t \to 0$ and $\tau(t) \to \infty$. No convergence assumption on the selected tuples P_t is needed for this conclusion. If in addition $P_t \to P$ and $\theta_t \to \theta$ within this rate class, then for each fixed $\tau > 0$ the component limit is $H_\theta(\tau; P)$. Its infinite-τ limit is I, and the corresponding iterated limits commute.*

Proof. The selected tuples lie in a common bounded set, the all-time error in (64) tends to zero, and $s_*/L \to 0$. Uniform effective-flow decay gives the diagonal assertion. Under the added phase and parameter convergence assumptions, finite-dimensional continuous dependence identifies the limit at each positive τ; entry has then occurred for all small t. The remaining assertion follows from effective-flow decay and fixed-fibre convergence. □

A general spatial initial metric at $\tau = 0$ need not equal its selected phase. Nor need a nonconvergent family of selected phases have any fixed-τ degeneration limit. Finally, part 3 of Proposition 8.3 shows failure of uniform relaxation to I near reducibility. It does not disprove tracking by a parameter-dependent effective trajectory, whose own relaxation time also diverges there.

9. Further questions

The weakest linear time scale in (74) is L/ϑ^2. Our relative spectral estimate resolves this scale without restricting the rate at which ϑ vanishes. The nonlinear residual bound (33), integrated by the trace comparison, does not by itself give tracking over every such interval. A proof at that scale would require a sharper residual, an additional cancellation in its weak projection, or a corrected effective equation with a suitable stability argument. Failure of the present accumulated bound is not a counterexample to that stronger tracking statement.

The four node lengths in this construction come from one genuine elliptic smoothing parameter. A theory with independently variable node capacities must first realize those parameters in a compact family and preserve the involution and bundle descent. In this quotient, compatibility pairs reflected nodes. An arbitrary list of lengths inserted into the effective energy would not establish that geometric realization.

The prescribed product-induced Kähler metric is essential to the exact K3 decomposition and the transverse-contraction argument. Replacing it by the Ricci-flat metric on each smooth threefold is a different analytic problem. Threefolds with full SU(3) holonomy likewise require new local bundle and spectral constructions; the finite product cover used here would no longer be available.

The inverse theorem recovers projective transport in its stated family. Recovering integral cohomological attachments, torsion, or intersection forms requires additional geometrically accessible integral data and a separate reconstruction theorem. The charge 48 in this paper is a directly computed integral pairing, not a quantity inferred from five real eigenvalues. Similarly, the affine metric on the effective tuple space is a metric on fibre metrics; no quotient metric on the threefold bundle-moduli space has been constructed here.

Finally, the unitary parameter locus has a locally free bundle and an explicit Hermitian–Einstein reference. It approaches a strictly polystable bundle without a singular graded quotient. Nonunitary instability, curvature concentration and singular graded objects would require new local models and matching estimates. Neither their defect scales nor their amplitude laws follow from the present smooth polystable degeneration.

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